DATABASE DESIGN

3D Asset Marketplace Platform

Version: 2.0

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1. Introduction

The database is designed using a **relational database architecture** to provide a secure, scalable, and maintainable foundation for the 3D Model Marketplace Platform. The platform manages three primary business modules:

- **3D Asset Marketplace**
- **Plugin Marketplace**
- **Custom Project Portal**

The database is designed to support more than **20,000 digital assets**, thousands of international customers, secure payment processing, digital downloads, and future expansion without major structural modifications.

The selected Database Management System (DBMS) is **PostgreSQL** because of its scalability, reliability, advanced indexing capabilities, JSON support, and excellent integration with Django.

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2. Database Objectives

The database is designed to:

- Store customer information securely.
- Manage thousands of 3D assets.
- Manage plugin products.
- Handle custom project requests.
- Store multiple downloadable file versions.
- Process international orders.
- Maintain secure payment records.
- Track digital downloads.
- Maintain customer reviews.
- Store project communications.
- Support future platform expansion.

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3. Main Database Modules

The database consists of the following modules:

- Authentication
- Marketplace
- Plugin Store
- Custom Project Portal
- Orders
- Payments
- Downloads
- Customer Dashboard
- Admin Dashboard
- Analytics

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4. Main Database Tables

User Management

- Users
- Roles
- User Profiles

Marketplace

- Categories
- Products
- Product Images
- Product Files
- Product Tags

Plugin Marketplace

- Plugins
- Plugin Categories
- Plugin Files
- Plugin Images

Custom Project Portal

- Custom Projects
- Project Files
- Project Messages
- Project Quotations
- Project Deliveries

Commerce

- Shopping Cart
- Cart Items
- Orders
- Order Items
- Payments
- Invoices
- Coupons

Customer Features

- Wishlist
- Reviews
- Notifications
- Download History

Administration

- Analytics
- Activity Logs

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5. Table Descriptions

Users

Stores registered customer and administrator accounts.

Fields

- id (UUID)
- full_name
- email
- password
- phone
- country
- avatar
- role_id
- email_verified
- is_active

- last_login
- created_at
- updated_at

Roles

Stores user roles.

Fields

- id
- role_name
- description

Default Roles

- Administrator
- Customer

User Profiles

Stores additional profile information.

Fields

- id
- user_id
- company_name
- website
- address
- city
- state
- postal_code
- country

Categories

Stores product categories.

Fields

- id
- name

- slug
- description
- image
- parent_category
- created_at

Examples:

- Architecture
- Furniture
- Interior
- Vehicles
- Characters
- Nature
- Electronics

Products

Stores all 3D models available for sale.

Fields

- id
- category_id
- name
- slug
- description
- short_description
- price
- currency
- discount_price
- thumbnail
- preview_model
- preview_video
- polygon_count
- vertex_count
- software_used
- animated
- rigged
- pbr_ready
- game_ready
- license_type
- average_rating
- total_downloads
- status
- created_at
- updated_at

Product Images

Stores gallery images.

Fields

- id
- product_id
- image_url
- display_order

Product Files

Stores downloadable product files.

Fields

- id
- product_id
- file_name
- file_format
- file_size
- storage_path
- version
- uploaded_at

Supported formats:

- .blend
- .fbx
- .obj
- .glb
- .gltf
- .max
- Textures
- HDRI
- ZIP

Product Tags

Stores searchable keywords.

Fields

- id

- product_id
- tag_name

Examples:

- Low Poly
- High Poly
- Game Ready
- PBR
- Animated

Plugin Categories

Stores plugin categories.

Examples:

- Blender
- Maya
- 3ds Max
- AutoCAD
- SketchUp

Plugins

Stores plugin products.

Fields

- id
- category_id
- name
- slug
- description
- version
- compatible_software
- price
- currency
- thumbnail
- documentation
- status
- created_at
- updated_at

Plugin Files

Stores downloadable plugin packages.

Fields

- id
- plugin_id
- version
- file_name
- file_size
- storage_path

Plugin Images

Stores plugin screenshots.

Fields

- id
- plugin_id
- image_url
- display_order

Shopping Cart

Stores customer shopping carts.

Fields

- id
- user_id
- created_at

Cart Items

Stores products added to the shopping cart.

Fields

- id
- cart_id

- product_type
- product_id
- quantity
- price

Product Type

- 3D Model
- Plugin

Orders

Stores customer purchases.

Fields

- id
- user_id
- invoice_number
- total_amount
- currency
- payment_status
- order_status
- created_at

Order Items

Stores purchased items.

Fields

- id
- order_id
- product_type
- product_id
- quantity
- purchase_price

Payments

Stores payment transactions.

Fields

- id
- order_id
- gateway
- transaction_id
- amount
- currency
- payment_status
- payment_date

Supported gateways:

- Stripe
- PayPal
- Razorpay

Invoices

Stores invoice information.

Fields

- id
- order_id
- invoice_number
- invoice_pdf
- created_at

Coupons

Stores promotional offers.

Fields

- id
- code
- discount_type
- discount_value
- expiry_date
- active

Downloads

Stores customer download records.

Fields

- id
- user_id
- product_type
- product_id
- signed_url
- expires_at
- download_count
- downloaded_at

Wishlist

Stores customer favorite products.

Fields

- id
- user_id
- product_type
- product_id

Reviews

Stores customer ratings.

Fields

- id
- user_id
- product_type
- product_id
- rating
- review
- created_at

Notifications

Stores platform notifications.

Fields

- id

- user_id
- title
- message
- read_status
- created_at

Custom Projects

Stores custom modeling requests submitted by customers.

Fields

- id
- customer_id
- project_title
- project_description
- category
- estimated_budget
- deadline
- project_status
- quotation_amount
- assigned_designer
- payment_status
- created_at
- updated_at

Project Files

Stores customer reference files and final project files.

Fields

- id
- project_id
- uploaded_by
- file_type
- file_name
- storage_path
- uploaded_at

File Types

- Reference Image
- CAD Drawing
- Sketch
- Final Model
- Texture
- ZIP

Project Messages

Stores communication between customer and administrator.

Fields

- id
- project_id
- sender_id
- message
- attachment
- created_at

Project Quotations

Stores project quotations.

Fields

- id
- project_id
- quotation_amount
- estimated_delivery
- quotation_status
- created_at

Quotation Status

- Pending
- Accepted
- Rejected

Project Deliveries

Stores completed project files.

Fields

- id
- project_id
- delivery_version
- delivery_notes
- storage_path

- delivered_at

Analytics

Stores business analytics.

Examples:

- Monthly Sales
- Revenue
- Downloads
- Best Selling Models
- Best Selling Plugins
- Active Customers

Activity Logs

Stores important platform activities.

Examples:

- Login
- Payment
- Product Upload
- Plugin Upload
- Download
- Project Submission
- Project Delivery

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6. Entity Relationships

- One Role → Many Users
- One User → One User Profile
- One User → Many Orders
- One User → Many Downloads
- One User → Many Reviews
- One User → Many Wishlist Items
- One Category → Many Products
- One Product → Many Images
- One Product → Many Files
- One Plugin Category → Many Plugins
- One Plugin → Many Files
- One Plugin → Many Images
- One Cart → Many Cart Items
- One Order → Many Order Items

- One Order → One Payment
- One Order → One Invoice
- One User → Many Custom Projects
- One Custom Project → Many Files
- One Custom Project → Many Messages
- One Custom Project → One Quotation
- One Custom Project → Many Deliveries

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7. Database Normalization

The database follows the **Third Normal Form (3NF)** to reduce redundancy and improve consistency.

Benefits include:

- Elimination of duplicate data
- Better storage efficiency
- Easier maintenance
- Improved query performance
- Higher data integrity

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8. Indexing Strategy

Indexes should be created on:

- Product Name
- Plugin Name
- Category
- Price
- Slug
- Email
- Transaction ID
- Invoice Number
- Project Status
- Created Date

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9. Data Security

The database follows industry-standard security practices:

- Password hashing using Django's built-in password hashing
- UUID primary keys
- Foreign key constraints
- Role-Based Access Control (RBAC)

- Secure payment references
- No storage of payment card details
- Cloud storage references instead of local file paths
- Automatic timestamps for auditing
- Signed download URLs
- Server-side payment verification

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10. Scalability

The database is designed to support:

- More than 20,000 marketplace assets
- More than 10,000 plugins
- Thousands of concurrent users
- Unlimited customer project requests
- Future AI-assisted services
- Multi-language support
- Multi-currency support
- Additional payment gateways
- Future seller marketplace functionality

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11. Conclusion

The proposed database architecture provides a secure, scalable, and maintainable foundation for the 3D Model Marketplace Platform. It supports three primary business modules—Marketplace, Plugin Store, and Custom Project Portal—while ensuring efficient management of users, digital assets, orders, payments, secure downloads, and project workflows. The modular design enables the platform to grow with future business requirements without requiring major structural redesign.